Division 3 - Metal Coils

Application

48 This Division applies to a flatbed vehicle or a sided vehicle or intermodal container that is transporting one or more metal coils that individually or grouped together have a total weight of 2 268 kilograms or more.

Coils transported with eyes vertical by a vehicle or an intermodal container with anchor points

49(1) This section applies to coils transported with the eyes vertical.

(2) Where a vehicle is transporting a single coil or several coils which are not grouped in a row, each coil shall be secured by tiedowns arranged in a manner that prevents the coil from tipping forward, rearward, or sideways. The securement system shall include:

(a) at least 1 tiedown attached diagonally from the left side of the vehicle near the forward-most part of the coil, across the eye of the coil, to the right side of the vehicle near the rearmost part of the coil,

(b) at least 1 tiedown attached diagonally from the right side of the vehicle near the forward-most part of the coil, across the eye of the coil, to the left side of the vehicle near the rearmost part of the coil,

(c) at least 1 tiedown attached across the eye of the coil, and

(d) blocking and bracing, friction mats or tiedowns that prevent the coil moving forward.

(3) Where a vehicle is transporting coils that are grouped and loaded side by side in a transverse or lengthwise row, each row shall be secured by

(a) at least 1 tiedown against the front of the row, restraining against forward movement, and where practical, making an angle of not more than 45 degrees with the deck when viewed from the side,

(b) at least 1 tiedown against the rear of the row, restraining against rearward movement, and where practical, making an angle of not more than 45 degrees with the deck when viewed from the side,

(c) at least one tiedown over the top of each coil or each transverse row of coils, restraining against vertical movement, and

(d) tiedowns shall be arranged to prevent shifting and tipping in the forward, rearward and lateral directions.

(4) Subject to subsections (2) and (3) a tiedown going over the top of a coil shall be as close as practical to the eye of the coil.

Coils transported with eyes crosswise by a vehicle or an intermodal container with anchor points

50(1) This section applies to coils transported with the eyes crosswise and to transverse rows of coils loaded side by side and having approximately the same outside diameters.

(2) Each coil or transverse row of coils shall be immobilized with timbers, chocks or wedges, a cradle or other device that

(a) prevents the coil from rolling,

(b) supports the coil off the deck, and

(c) is not capable of becoming unfastened or loose while the vehicle is on a highway.

(3) Where timbers, chocks or wedges are used to secure a coil, they shall be held in place by coil bunks or similar devices to prevent the blocking device from coming loose.

(4) Each coil or transverse row of coils shall be secured with

(a) at least 1 tiedown through its eye, restricting against forward movement, and where practical, making an angle of not more than 45 degrees with the deck when viewed from the side, and

(b) at least one tiedown through its eye, restricting against rearward movement, and where practical, making an angle of not more than 45 degrees with the deck when viewed from the side.

(c) if coils are loaded to contact each other in the longitudinal direction and relative motion between the coils and between coils and the vehicle is prevented by tiedowns

i. Only the foremost and rearmost coils or rows of coils must be immobilized per Section 50(2)

ii. A single tiedown restraining against forward motion may be used to secure any coil except the rearmost one, which must be restrained against rearward motion

Coils transported with eyes lengthwise by a vehicle or an intermodal container with anchor points

51 An individual metal coil transported with the eye lengthwise shall be secured in accordance with one of sections 52, 53 or 54.

52(1) Each coil shall be immobilized by timbers, chocks or wedges, a cradle or another method that

(a) prevents the coil from rolling,

(b) supports the coil off the deck, and

(c) is not capable of becoming unfastened or loose while the vehicle is on a highway.

(2) Where timbers, chocks or wedges are used to secure a coil, they shall be held in place by coil bunks or similar devices to prevent the blocking device from coming loose.

(3) Each coil shall be secured with

(a) at least 1 tiedown attached diagonally through its eye from the left side of the vehicle near the forward-most part of the coil, to the right side of the vehicle near the rearmost part of the coil, making an angle of not more than 45 degrees, where practical, with the deck when viewed from the side,

(b) at least 1 tiedown attached diagonally through its eye, from the right side of the vehicle near the forward-most part of the coil, to the left side of the vehicle near the rearmost part of the coil, making an angle of not more than 45 degrees, where practical, with the deck when

viewed from the side,

(c) at least 1 tiedown attached across the top of the coil, and

(d) blocking or friction mats to prevent a coil from moving lengthwise.

53(1) Each coil shall be immobilized by timbers, chocks or wedges, a cradle or another method that

(a) prevents the coil from rolling,

(b) supports the coil off the deck, and

(c) is not capable of becoming unfastened or loose while the vehicle is on a highway.

(2) Where timbers, chocks or wedges are used to secure a coil, they shall be held in place by coil bunks or similar devices to prevent the blocking device from coming loose.

(3) Each coil shall be secured with

(a) at least 1 tiedown attached straight through its eye from the left side of the vehicle near the forward-most part of the coil, to the left side of the vehicle near the rearmost part of the coil and, where practical, making an angle of not more than 45 degrees with the deck when viewed from the side,

(b) at least 1 tiedown attached straight through its eye, from the right side of the vehicle near the forward-most part of the coil, to the right side of the vehicle near the rearmost part of the coil and, where practical, making an angle of not more than 45 degrees with the deck when viewed from the side,

(c) at least 1 tiedown attached across the top of the coil, and

(d) blocking or friction mats to prevent the coil from moving lengthwise.

54(1) Each coil shall be immobilized by timbers, chocks or wedges, a cradle or another method that

(a) prevents the coil from rolling,

(b) supports the coil off the deck, and

(c) is not capable of becoming unfastened or loose while the vehicle is on a highway.

(2) Where timbers, chocks or wedges are used to secure a coil, they shall be held in place by coil bunks or similar devices to prevent the blocking device from coming loose.

(3) Each coil shall be secured by

(a) at least 1 tiedown over the top of the coil, located near the forward-most part of the coil,

(b) at least 1 tiedown over the top of the coil located near the rearmost part of the coil, and

(c) blocking or friction mats to prevent the coil from moving lengthwise.

Rows of Coils with Eyes Lengthwise

55(1) This section applies to the transportation of transverse rows of metal coils with eyes lengthwise and with approximately equal outside diameters.

(2) A transverse row of coils shall be immobilized by timbers, chocks or wedges, a cradle or another method that

(a) prevents the coils from rolling,

(b) supports the coils off the deck, and

(c) is not capable of becoming unfastened or loose while the vehicle is on a highway.

(3) Where timbers, chocks or wedges are used to secure a row of coils, they shall be held in place by coil bunks or similar devices to prevent the blocking device from coming loose.

(4) A transverse row of coils shall be secured by

(a) at least 1 tiedown over the top of each coil, located near the forward-most part of the coil,

(b) at least 1 tiedown over the top of each coil, located near the rearmost part of the coil, and

(c) blocking or friction mats to prevent each coil from moving lengthwise.

Prohibitions

56 The use of nailed wood blocking or cleats as the sole means to secure timbers, chocks or wedges, or a nailed wood cradle is prohibited when metal coils are transported with eyes lengthwise or eyes crosswise by a vehicle or an intermodal container with anchor points.

57 When coils are transported with eyes crosswise, attaching tiedowns diagonally through the eye of a coil to form an X pattern when viewed from above the vehicle is prohibited.

Securement of Coils Transported in a Sided Vehicle or Intermodal Container without Anchor Points

58 Metal coils shall be secured in a manner to prevent shifting and tipping using a system of blocking and bracing, friction mats, tiedowns, or a combination of these.

Division 4 - Paper Rolls

Application

59(1) This Division applies to the transportation of paper rolls which individually or together weigh 2 268 kilograms or more.

(2) Paper rolls may be secured in accordance with this Division where:

(a) the total weight is less than 2 268 kilograms, and

(b) a single or several paper rolls are unitized on a pallet.

Friction mats

60 A friction mat used to provide the principal securement for a paper roll shall protrude from beneath the roll in the direction in which it is providing that securement.

Chocks, wedges and blocking

61 Chocks, wedges or blocking used to secure paper rolls shall not shift or become unfastened while the vehicle is on a highway.

Banding

62 Where paper rolls are banded together,

(a) the rolls shall be placed tightly against each other to form a stable group,

(b) the banding shall be applied tightly and remain so, and

(c) the banding shall be secured so that it cannot fall off the rolls or to the deck.

Single layer of paper rolls transported eyes vertical in a sided vehicle

63(1) This section applies to paper rolls that are transported with the eyes vertical in a single layer in a sided vehicle.

(2) The paper rolls shall be placed tightly against the front and both side walls of the vehicle, other paper rolls or other cargo.

(3) Where there are not enough paper rolls in a group of paper rolls to reach the walls of the vehicle, void fillers, blocking, bracing, tiedowns or friction mats shall be used to prevent the rolls moving sideways.

(4) Paper rolls may be banded together.

(5) Rearward movement shall be prevented by blocking, bracing, tiedowns or friction mats or by banding to other rolls.

(6) A paper roll shall be prevented from tipping or falling sideways or rearwards by banding it to other rolls or by using bracing or tiedowns where

(a) the vehicle's structure or other cargo does not prevent the roll from tipping or falling sideways or rearwards, and

(b) the width of the roll is more than 2 times its diameter.

(7) A single paper roll or the forward most roll in a group of paper rolls shall be prevented from tipping or falling forward by banding it to other rolls or by using bracing or tiedowns where

(a) the vehicle's structure or other cargo does not prevent the roll from tipping or falling forward,

(b) the roll is restrained against moving forward only by friction mats, and

(c) the width of the roll is more than 1.75 times its diameter.

(8) A single paper roll or the forward most roll in a group of paper rolls shall be prevented from tipping or falling forward by banding it to other rolls or by using bracing or tiedowns where

(a) the vehicle's structure or other cargo does not prevent the roll from tipping or falling forward, and

(b) the roll is not restrained against moving forward by friction mats, and

(c) the width of the roll is more than 1.25 times its diameter

Split cargo of paper rolls transported with eyes vertical in a sided vehicle

64(1) This section applies to split cargo of paper rolls transported with the eyes vertical in a sided vehicle

(2) Where a paper roll in a split cargo of paper rolls transported with the eyes vertical in a sided vehicle is not prevented from moving forward by the vehicle's structure or other cargo, it shall be prevented from moving forward

(a) by filling the open space, or

(b) by using blocking, bracing, tiedowns, friction mats as described in Section 63, or

(c) by using a combination of the methods in clauses (a) and (b).

Stacked cargo of paper rolls transported with eyes vertical in a sided vehicle

65(1) This section applies to stacked cargo of paper rolls transported with the eyes vertical in a sided vehicle.

(2) Section 63 applies to the bottom layer in a stacked cargo of paper rolls.

(3) Paper rolls shall not be loaded on top of another layer unless the layer beneath extends to the front of the vehicle.

(4) Paper rolls in the 2nd and subsequent layers shall be prevented from moving forward, rearward or sideways

(a) in accordance with section 63, or

(b) by using a blocking roll from a lower layer.

(5) A blocking roll used to prevent forward, rearward or sideways movement shall be

(a) at least 38 millimetres taller than other rolls, or

(b) raised at least 38 millimetres using dunnage or another method.

(6) Despite subsection (5), a roll at the rear end of a layer of rolls shall not be raised using dunnage.

Single layer of paper rolls transported eyes crosswise in a sided vehicle

66(1) This section applies to a single layer of paper rolls transported with the eyes crosswise in a sided vehicle.

(2) The paper rolls shall be prevented from rolling or moving lengthwise

(a) by contact with the vehicle's structure or other cargo, or

(b) by chocks, wedges, blocking, bracing or tiedowns.

(3) Void fillers, blocking, bracing, friction mats or tiedowns shall be used to prevent a paper roll or group of rolls from moving sideways towards the side walls of the vehicle

(a) where the total void space between the ends of a paper roll and the walls of the vehicle is more than 203 millimetres, or

(b) where the total void space in a row of paper rolls between the vehicle walls is more than 203 millimetres

Rear doors

67 A vehicle transporting paper rolls with the eyes crosswise shall not use the rear doors of the vehicle

(a) to secure the rearmost paper roll or layer of paper rolls, or

(b) to hold blocking that secures the rearmost paper roll or layer of paper rolls.

Stacked cargo of paper rolls transported eyes crosswise in a sided vehicle

68(1) This section applies to stacked cargo of paper rolls transported in a sided vehicle with the eyes crosswise.

(2) Section 66 applies to the bottom layer in a stacked cargo of paper rolls.

(3) Paper rolls shall not be loaded in a 2nd layer unless the bottom layer extends to the front of the vehicle.

(4) Paper rolls shall not be loaded in a 3rd or higher layer unless all the cylinder wells in the layer beneath are filled.

(5) The foremost paper roll in each upper layer and a roll with an empty cylinder well in front of it shall be secured against moving forward by

(a) banding it to other paper rolls, or

(b) blocking against a secured eye-vertical blocking roll resting on the deck that is at least 1.5 times taller than the diameter of the roll being blocked, or

(c) placing it in a cylinder well formed by 2 paper rolls on the lower layer with a diameter equal

to or greater than that of the paper roll on the upper layer.

(6) The rearmost paper roll in each upper layer shall be secured by banding it to other paper rolls where it is located in either of the last 2 cylinder wells formed by the rearmost paper rolls in the layer below.

(7) Void fillers, blocking, bracing, friction mats or tiedowns shall be used to prevent a paper roll or group of rolls from moving sideways towards the side walls of the vehicle

(a) where the total void space between the ends of a paper roll and the walls of the vehicle is more than 203 millimetres, or

(b) where the total void space in a row of paper rolls between the vehicle walls is more than 203 millimetres

Single layer of paper rolls transported eyes lengthwise in a sided vehicle

69(1) This section applies to paper rolls transported in a sided vehicle in a single layer with the eyes lengthwise.

(2) A paper roll shall be prevented from moving forward by contact with the vehicle's structure or other cargo or by blocking or tiedowns.

(3) A paper roll shall be prevented from moving rearward by contact with other cargo or by blocking, friction mats, tiedowns or cradles with friction mats applied between the roll and the cradle and between the cradle and the floor.

(4) A paper roll shall be prevented from rolling or moving sideways by contact with the vehicle's wall or other cargo or by chocks, wedges, cradles or other blocking mechanism.

(5) Cradles used to support and restrain paper rolls must be secured against movement by using chocks, wedges and blocking, or friction mats.

(a) The width of a cradle(s) used to support a roll must be:

(i) at least ½ times the height of the roll, as measured from the lowest point on the roll, or

(ii) the roll must be attached to the cradle with bands or straps.

(b) When used to secure cargo in the lateral direction, each cradle:

(i) must contact at least 1/8 of the roll's perimeter, or

(ii) must be attached to the roll with bands or straps.

Stacked cargo of paper rolls transported eyes lengthwise in a sided vehicle

70(1) This section applies to stacked cargo of paper rolls transported with the eyes lengthwise in a sided vehicle

(2) Section 69 applies to the bottom layer in a stacked cargo of paper rolls.

(3) Paper rolls shall not be loaded in a higher layer unless all the cylinder wells in the layer beneath are filled.

(4) An upper layer of paper rolls shall be formed by placing the paper rolls in the cylinder wells formed by the rolls beneath.

(5) Paper rolls in the 2nd and higher layers shall be prevented from moving forward or rearward

(a) in accordance with section 69, or

(b) by using a blocking roll from a lower layer, or

(c) by banding them to other rolls.

Cargo of Paper rolls transported with eyes vertical or with eyes lengthwise on a flatbed vehicle or in a curtain sided vehicle

71(1) This section applies to cargo of paper rolls transported with the eyes vertical or with eyes lengthwise on a flatbed vehicle or in a curtain sided vehicle

(2) The paper rolls shall be loaded and secured as described for a sided vehicle, and the entire load shall be secured by tiedowns in accordance with the provisions of section 22 of this Standard.

(3) Stacked loads of paper rolls with eyes vertical are prohibited.

Cargo of Paper rolls transported with eyes crosswise on a flatbed vehicle or in a curtain sided vehicle

72(1) This section applies to cargo of paper rolls transported with the eyes crosswise on a flatbed vehicle or in a curtain sided vehicle

(2) The paper rolls shall be prevented from rolling or shifting longitudinally by contact with vehicle structure or other cargo, by chocks, by wedges, by blocking and bracing of adequate size, or by tiedowns.

(3) When used, chocks, wedges or blocking must be held securely in place by some means in addition to friction so that they cannot become unfastened or loose while the vehicle is on a highway.

(4) Tiedowns shall be used in accordance with the provisions of section 22 of this Standard to prevent lateral movement.

Division 5 -Concrete Pipe

In this Division pipe means concrete pipe.

Application

73(1) This Division applies to the transportation of concrete pipes loaded transversely on a flatbed vehicle

(2) This Division does not apply to

(a) unitized concrete pipes that has no tendency to roll, or

(b) concrete pipe loaded in a sided vehicle.

(3) Section 22 does not apply to the transportation of concrete pipes to which this Division applies.

Tiedowns

74 A transverse tiedown running through a pipe in an upper layer or over lengthwise tiedowns is considered to secure all the concrete pipe in a lower layer on which that tiedown causes pressure.

Blocking

75(1) The blocking used to secure a pipe shall comply with this section.

(2) Blocking shall be placed symmetrically about the centre of the pipe

(3) Where one piece of blocking is used it shall extend to at least ½ the distance from the centre to each end of the pipe.

(4) Where two pieces of blocking are used they shall be placed near each end of the pipe

(5) Blocking shall be placed firmly against a pipe and shall be secured to prevent it moving out from under the pipe

(6) Timber blocking shall have a minimum dimension of at least 8.9 centimetres by 14 centimetres

Multi-sized pipe

76 Where the pipes to be transported on a vehicle have different diameters,

(a) pipes with the same diameter shall be grouped together, and

(b) each group shall be secured separately.

Arranging layers

77(1) This section applies to pipe arranged in layers on a vehicle.

(2) The bottom layer of pipes shall be arranged to cover the full length of the vehicle or, as a partial layer with one or 2 groups.

(3) Pipe in an upper layer shall be placed only in the cylinder wells formed by adjacent pipes in the layer beneath.

(4) A 3rd or higher layer of pipe shall not be formed unless all the cylinder wells in the layer beneath are filled with pipe.

(5) The top layer shall be arranged as a complete layer or, as a partial layer in one or 2 groups.

Bell Pipe

78(1) Bell pipe shall be loaded on at least 2 lengthwise spacers of sufficient height so that the bell is clear of the deck.

(2) Bell pipes loaded in a single layer shall have the bells alternating on opposite sides of the vehicle.

(3) The ends of consecutive bell pipes shall be

(a) staggered, where possible, within the allowable width, or

(b) aligned.

(4) Stacked bell pipes shall have all the bells of the bottom layer on the same side of the vehicle.

(5) Stacked bell pipes shall be loaded with bells on the opposite side of the vehicle to the bells of the pipe in the layer beneath.

(6) Where the second layer of stacked bell pipes is not complete, the bells of the pipes in the bottom layer that do not support a pipe above shall alternate on opposite sides of the vehicle.

Single layer or Bottom layer, small pipe

79(1) This section applies to small pipes arranged in a single layer or in the bottom layer of stacked pipes.

(2) The front and the rear pipes of each group of pipes arranged in a single layer or in bottom layer shall be immobilized lengthwise at each end by blocking, vehicle end structure, stakes, a locked pipe unloader or other equivalent means.

(3) Pipe that is not at the end of a group of pipes arranged in a single layer or in a bottom layer may be held in place by blocks, wedges or both.

(4) A pipe in a single layer or in the bottom layer shall be held firmly in contact with the adjacent pipe by tiedowns running through the front and rear pipes of a group of pipes:

(a) One or more tiedowns running through the front pipe of each group of pipes arranged in a single layer or in a bottom layer shall run rearward at an angle, where practical, of not more than 45 degrees with the horizontal, and

(b) One or more tiedowns running through the rear pipe of each group of pipes arranged in a single layer or in a bottom layer shall run forward at an angle, where practical, of not more than 45 degrees with the horizontal.

Tiedowns for securing layers of small concrete pipes

80(1) This section applies to all cargo of small pipes.

(2) Each pipe shall be secured with a tiedown running through it, or

(3) Where each pipe is not secured individually with a tiedown

(a) one 1.27 centimetre (1/2 inch) diameter chain or wire rope or two 0.95 centimetre (3/8 inch) diameter chains or wire ropes shall be placed lengthwise over the group of pipes, and

(b) one transverse tiedown shall be used for every 3.04 metres of cargo length.

(4) The transverse tiedowns referred to in subsection (3) shall be placed

(a) through a pipe on the top layer, or

(b) over the lengthwise tiedown between 2 pipes on the top layer.

Top layer, small pipes

81(1) This section applies to small pipes in the top layer of stacked pipes.

(2) Where the first pipe of a group in the top layer is not placed in the first cylinder well formed by the pipes at the front of the layer beneath, it shall be secured by an additional tiedown that

(a) runs rearward at an angle, where practical, of not more than 45 degrees to the horizontal, and

(b) passes through the front pipe of the upper layer or outside before the front pipe of the upper layer and over the lengthwise tiedown

(3) Where the last pipe of a group in the top layer is not placed in the last cylinder well formed by the pipes at the rear of the layer beneath, it shall be secured by an additional tiedown that

(a) runs forward at an angle, where practical, of not more than 45 degrees to the horizontal, and

(b) passes through the rear pipe of the upper layer or outside after the rear pipe of the upper layer and over the lengthwise tiedown.

Large pipes

82(1) This section applies to the transportation of large pipe.

(2) The front pipe and the rear pipe in a group of pipes transported on a vehicle shall be immobilized by blocking, wedges, vehicle end structure, stakes, locked pipe unloader or other equivalent means

(3) Each pipe in the front half of the group of pipes, including the middle one where there is an odd number, shall have at least one tiedown that passes through the pipe

(a) running rearward at an angle, where practical, of not more than 45 degrees with the horizontal, and

(b) holding the pipe firmly in contact with an adjacent pipe.

(4) Each pipe in the rear half of the group of pipes, shall have at least one tiedown that passes through the pipe

(a) running forward at an angle, where practical, of not more than 45 degrees with the horizontal, and

(b) holding the pipe firmly in contact with an adjacent pipe.

(5) Where the front pipe in a group of pipes is not in contact with the vehicle's end structure, stakes or other equivalent means, it shall be secured by at least 2 tiedowns positioned in accordance with subsections (3).

(6) Where the rear pipe in a group of pipes is not in contact with the vehicle's end structure, stakes, a locked pipe unloader, or other equivalent means, it shall be secured by at least 2 tiedowns positioned in accordance with subsections (4).

(7) Where the vehicle is transporting a single pipe, or several pipes that do not touch each other, a pipe shall be secured under this section as where it were the front or rear pipe in a group of pipes.

Division 6 - Intermodal Containers

Application

83 This Division applies to the transportation of intermodal containers.

Intermodal container transported on Container chassis vehicle

84(1) This section applies to the transportation of an intermodal container on a container chassis vehicle

(2) Despite section 22, an intermodal container shall be secured to the container chassis with integral locking devices.

(3) The integral locking devices used shall restrain each lower corner of the intermodal container.

(a) *deleted January 2013*

(b) *deleted January 2013*

(c) *deleted January 2013*

(d) *deleted January 2013*

(e) *deleted January 2013*

(4) The front and the rear of the intermodal container shall be independently secured.

Loaded intermodal container transported on other vehicles

85(1) This section applies to the transportation of a loaded intermodal container on a vehicle that is not a container chassis vehicle.

(2) All the lower corners of a loaded intermodal container shall

(a) rest on the vehicle, or

(b) be supported by a structure

(i) capable of bearing the weight of the container, and

(ii) independently secured to the vehicle.

(3) Despite section 22, an intermodal container shall be secured to the vehicle by either or both

(a) chains, wire ropes or integral locking devices that are fixed to all the lower corners,

(b) crossed chains that are fixed to all the upper corners.

(4) The front and the rear of the intermodal container shall be independently secured.

Empty intermodal container transported on other vehicles

86(1) This section applies to the transportation of an empty intermodal container by a vehicle other than a container chassis vehicle.

(2) All the lower corners of an empty intermodal container shall

(a) rest on the vehicle, or

(b) be supported by a structure

(i) capable of bearing the weight of the container, and

(ii) independently secured to the vehicle.

(3) An empty intermodal container is not required to comply with subsection (2) where

(a) the container is balanced, positioned and stable on the vehicle before tiedowns or other securing devices are attached, and

(b) the container does not overhang either the front or rear of the vehicle by more than 1.5 metres

(4) An empty intermodal container shall not interfere with the vehicle's manoeuvrability.

(5) An empty intermodal container shall be secured against moving sideways, lengthwise or vertically in accordance with:

a) the provisions of section 85(3) and 85(4) of this Standard, or

b) the provisions of section 22 of this Standard.

Division 7 - Vehicles as Cargo

Application

87(1) This Division applies to the transportation of light vehicles, heavy vehicles and flattened or crushed light vehicles.

Light vehicles

88(1) Light vehicles shall be secured in accordance with this section.

(2) Despite section 22, a light vehicle shall be restrained at both the front and rear from moving sideways, forward, rearward and vertically using a minimum of 2 tiedowns.

(3) Tiedowns that are designed to attach to the structure of a light vehicle shall be attached to the mounting points on the vehicle that are specifically designed for that purpose.

(4) Tiedowns that are designed to fit over or around the wheels of a light vehicle shall restrain the vehicle from moving sideways, forward, rearward and vertically.

(5) Despite section 20, edge protectors are not required for synthetic webbing at points where the webbing comes into contact with the tires of the light vehicle.

(6) It is prohibited to transport stacks of light vehicles.

Heavy vehicles

89(1) Heavy vehicles shall be transported in accordance with this section.

(2) Accessory equipment on a heavy vehicle, including a hydraulic shovel, shall be completely lowered and secured to the vehicle unless:

(a) the accessory equipment can only move vertically;

(b) accessory equipment that can pivot, tilt or move sideways is blocked or immobilized by the transporting vehicle's structure or by a blocking or securement mechanism built into the transported vehicle.

(3) Articulated vehicles shall be restrained in a manner that prevents articulation while the vehicle is on a highway.

(4) Despite section 22, a heavy vehicle with crawler tracks or wheels shall be restrained against moving sideways, forward, rearward and vertically by at least 4 tiedowns,

(a) each with a working load limit of at least 2 268 kilograms, and

(b) each attached, as close as practical, at the front and rear of the vehicle or to mounting points on the vehicle that are specifically designed for that purpose.

Flattened or crushed light vehicles

90 Flattened or crushed light vehicles shall be secured in accordance with the sections 91, 92 and 93.

Prohibition

91 (1) Synthetic webbing shall not be used to secure flattened or crushed light vehicles

(2) Despite Section 91(1) synthetic webbing may be used to connect wire rope or chain to anchor points on the transporting vehicle where the webbing is no more than 15 cm above the deck of the vehicle and must not come in contact with the flattened or crushed vehicles.

Securement system for immobilizing stacks of flattened or crushed vehicles

92(1) Flattened or crushed light vehicles shall be transported with vehicles which.

(a) have containment walls or comparable structures on 4 sides that

(i) extend to the full height of the cargo, and

(ii) prevent the cargo moving forward, rearward and sideways, or

(b) have containment walls or comparable structures on 3 sides that

(i) extend to the full height of the cargo, and

(ii) prevent the cargo moving forward, rearward and to one side, and

(iii) have 2 or more tiedowns per stack of flattened or crushed vehicles, or

(c) have containment walls or comparable structures on the front and rear that

(i) extend to the full height of the cargo, and

(ii) prevent the cargo moving forward and rearward, and

(iii) have 3 or more tiedowns per stack of flattened or crushed vehicles, or

(d) have 4 or more tiedowns per stack of flattened or crushed vehicles.

(2) Each tiedown referred to in subsection 1 shall have a working load limit of 2 268 kilograms or more.

Containment of Loose Parts

93(1) A vehicle transporting flattened or crushed light vehicles shall have equipment that

(a) extends the full height of the cargo, and

(b) prevents any loose part of the load from falling from the vehicle.

(2) For the purposes of subsection (1), the equipment used to contain loose parts shall be structural walls, floors, sides or sideboards or suitable covering material, alone or in combination.

Division 8 - Roll-on/roll-off and Hook Lift Containers

Application

94 This Division applies to the transportation of roll-on/roll-off containers and hook lift containers.

Replacement of Securing Devices

95(1) Where a front stop or lifting device of a securement system on a vehicle that is not equipped with an integral securement system is missing, damaged or not compatible with the securing devices on a container, additional manually installed tiedowns shall be used to secure the container to the vehicle.

(2) Where a front stop or lifting device of an integral securement system on a vehicle is missing, damaged or not compatible with the securing devices on a container, the container shall be secured to the vehicle using manually installed tiedowns.

(3) A manually installed tiedown shall provide the same level of securement as the missing, damaged or incompatible component it replaces.

No integral securement system

96(1) A roll-on/roll-off container or hook lift container transported by a vehicle that is not equipped with an integral securement system

(a) shall be blocked against forward movement

(i) by the lifting device acting as a blocking structure, or

(ii) at least two stops located approximately the same distance from the longitudinal axis of the container, or

(iii) a combination of (i) and (ii)

(b) shall be secured to the front of the vehicle

(i) by the lifting device, or

(ii) by another securing device which restrains against sideways and vertical movement, and

(c) shall be secured to the rear of the vehicle in accordance with at least one of the following:

(i) one tiedown attached to the vehicle chassis and to both sides of the container;

(ii) 2 tiedowns installed lengthwise, each securing one side of the container to one of the vehicle's side rails;

(iii) 2 hooks, or equivalent mechanisms, securing both sides of the container to the vehicle chassis at least as effectively as the tiedowns referred to in subclauses (i) and (ii).

(2) A device used to secure a roll-on/roll off or hook lift container to the rear of a vehicle that is not equipped with an integral securement system

(a) shall be installed not more than 2 metres from the rear of the container, and

(b) all tiedowns shall have a working load limit of at least 2 268 kilograms.

(3) Sections 10(2) and 10(3) do not apply to this Division.

Division 9 - Boulders

Application

97(1) This Division applies to the transportation of boulders

(a) on a flatbed vehicle, or

(b) in a vehicle whose sides are not designed and rated to contain such a cargo.

(2) A piece of natural, irregularly shaped rock that weighs more than 100 kilograms but less than 5000 kilograms may be secured in accordance with this Division.

(3) A piece of natural, irregularly shaped rock of any size may be contained within a vehicle that is designed to carry such a cargo.

(4) A piece of rock of any size that is artificially formed or cut into shape and has a stable base for securement may be secured in accordance with this Division.

Positioning on vehicle

98(1) A boulder shall be placed with its flattest or its largest side down

(2) A boulder shall be supported on at least 2 pieces of hardwood blocking

(a) with side dimensions of not less than 8.9 centimetres by 8.9 centimetres

(b) that extend the full width of the boulder,

(c) that are placed as symmetrically as possible under the boulder, and

(d) that support at least 3/4 of the length of the boulder.

(3) Where the flattest side of a boulder is rounded or partially rounded and the boulder may roll,

(a) the boulder shall be in a crib made of hardwood fixed to the deck,

(b) the boulder shall rest on both the deck and the crib, and

(c) the boulder shall have at least three well-separated points of contact with the crib and deck to prevent the boulder from rolling in any direction.

(4) Where a boulder is tapered, the narrowest end shall point towards the front of the vehicle.

Tiedowns

99(1) A tiedown used to secure a boulder shall be made of chain.

(2) A tiedown that touches a boulder

(a) shall, where possible, be located in valleys or notches across the top of the boulder, and

(b) shall be arranged so that it does not slide across the rock surface.

Number of tiedowns

100(1) Despite section 22, each cubic shaped boulder shall be secured with 2 or more tiedowns placed

(a) transversely across the vehicle, and

(b) as close as possible to the hardwood blocking.

(2) Despite section 22, each non-cubic shaped boulder with a stable base shall be secured with 2 or more tiedowns

(a) forming an "X" pattern over the boulder,

(b) passing over the centre of the boulder, and

(c) attached to each other, where they intersect, by a shackle or other connecting device.

(3) Despite sections 10(2), 10(3) and 22, each non-cubic shaped boulder with unstable base shall be secured with

(a) one tiedown surrounding the top of the boulder

(i) located at a point between 1/2 and 2/3 of the height of the boulder, and

(ii) having a working load limit of at least the half of the weight of the boulder, and

(b) 4 tiedowns, each

(i) attached to the surrounding tiedown and to the vehicle that prevent the boulder moving horizontally, and

(ii) having a working load limit of at least 1/4 the weight of the boulder, and

(iii) shall be placed at an angle, where practical, of not more than 45 degrees from the horizontal

PART 3 - Default Working Load Limits

Deleted September 2010

PART 4 - Manufacturing Standards

Section 1 - Vehicle Structure

Truck Trailer Manufacturers Association - RP 47

Section 2 - Anchor Points

Canadian Motor Vehicle Safety Standard (CMVSS 905)

Truck Trailer Manufacturers Association - RP47

Section 3 - Platform Bodies (Flatdecks)

Truck Trailer Manufacturers Association - RP47

Section 4 - Van, Sided & Dump Bodies

Truck Trailer Manufacturers Association - RP47

Web Sling and Tiedown Association

Recommended Standard Specification for Interior Van Securement WSTDA-T5

Section 5 - Tiedowns

Web Sling and Tiedown Association

Recommended Standard Specification for Synthetic Webbing Tiedowns WSTDA-T1

Recommended Standard Specification for Winches Used With Synthetic Web Tiedowns WSTDA-T3

Recommended Standard Specification for Interior Van Securement WSTDA-T5

Section 6 - Webbing Assemblies

Web Sling and Tiedown Association

Recommended Standard Specification for Synthetic Webbing Tiedowns WSTDA-T1

Recommended Operating, Care and Inspection Manual for Synthetic Web Tiedowns WSTDA-T2

Recommended Standard Specification for Synthetic Webbing Used for Tiedowns WSTDA-T4

Section 7 - Chain Assemblies

National Association of Chain Manufacturers

Welded Steel Chain Specifications

Welded Steel Chain – Working Load Limits

Size mm (in)	Grade 3 Proof coil	Grade 43 High test	Grade 70 Transport	Grade 80 Alloy	Grade 100 Alloy
7 mm (1/4 in)	580 kg (1300 lb.)	1180 kg (2600 lb)	1430 kg (3150 lb)	1570 kg (3500 lb)	1950 kg (4300 lb)
8 mm (5/16 in)	860 kg (1900 lb.)	1770 kg (3900 lb)	2130 kg (4700 lb)	2000 kg (4500 lb)	2600 kg (5700 lb)
10 mm (3/8 in)	1200 kg (2650 lb.)	2450 kg (5400 lb)	2990 kg (6600 lb)	3200 kg (7100 lb)	4000 kg (8800 lb)
11 mm (7/16 in)	1680 kg (3700 lb.)	3270 kg (7200 lb)	3970 kg (8750 lb)	-	-
13 mm (1/2 in)	2030 kg (4500 lb.)	4170 kg (9200 lb)	5130 kg (11300 lb)	5400 kg (12000 lb)	6800 kg (15000 lb)
16 mm (5/8 in)	3130 kg (6900 lb.)	5910 kg (13000 lb)	7170 kg (15800 lb)	8200 kg (18100 lb)	10300 kg (22600 lb)
Chain Marks	3 30 300	4 43 430	7 70 700	8 80 800	10 100 1000

Section 8 - Wire Rope and Attachments

Wire Rope Technical Board

Wire Rope Users Manual

Section 9 - Synthetic Rope and Attachments

Cordage Institute:

CI-1301-96 Polyester Fiber Rope, 3 and 8 Strand Constructions

CI-1302A-96 Polyester/Polyolefin Dual Fiber Rope, 3 Strand Construction

CI-1302B-99 Polyester/Polyolefin Dual Fiber Rope, 8 Strand Construction

CI-1304-96 Polyester Fiber Rope, 3 and 8 Strand Constructions

CI-1305-96 Single Braided Polyester Fiber Rope, 12 Strand Construction

CI-1307-96 Polyester Fiber Rope, Double Braid Construction

CI-1307-96 Polyester Fiber Rope, High Performance Double Braid Construction

CI-1303-96 Nylon (Polyamide) Fiber Rope, 3 and 8 Strand Constructions

CI-1307-96 Nylon (Polyamide) Fiber Rope, Double Strand Construction

CI-1307-96 Nylon (Polyamide) Fiber Rope, High Performance Double Braid Construction

Section 10 - Steel Strapping

American Society for Testing and Materials

Standard Specification for Strapping, Flat Steel and Seals (ASTM D3953-91)

Section 11 - Clamps and Latches

International Standards Organization - 668.

Section 12 - Roll-on/Roll-off Containers

American National Standards Institute

Mobile Wastes and Recyclable Materials Collection, Transportation, and Compaction Equipment - Safety Requirements (ASC Z245.1 -1999)

Waste Containers - Safety Requirements (ASC Z245.30 -1999)

Waste Containers - Compatibility Requirements (ASC Z245.60 -1999)